Eduardo Hernández Vergara

Computer Systems Engineer · Backend & Cloud · Go · AWS · Complex Systems & Cellular Automata

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PROFESSIONAL SUMMARY

Software engineer focused on backend and cloud. I designed and built, on my own, the AWS backend of an IoT platform for the power grid: Go microservices on SQS and NATS, deployed to Kubernetes and audited with CloudTrail, plus a metering-file pipeline with idempotent persistence and retries. I also moved about eight desktop apps to a single cross-platform codebase and wrote apps that talk to hardware over BLE and USB. I come from research in complex systems and cellular automata: an honors thesis and a CANDAR 2024 paper. Looking for backend, cloud, or software engineering roles, on-site, hybrid, or remote; also open to relocation.

WORK EXPERIENCE

Backend Developer

Distribuciones Cantilever Jul 2025 – Present

Owner of the cloud backend and apps for IoT metering and power-grid monitoring platforms, from AWS down to field devices.

- Designed and built the entire cloud backend: Go and Java microservices that consume AWS SQS queues and S3 and CloudTrail events, talk over NATS, and deploy to Kubernetes on Azure. AWS activity is audited through CloudTrail; today they move about 10,000 messages a day across several SQS and NATS queues.
- AWS administration: CloudTrail, EC2, S3, EKS, DynamoDB, EventBridge, SQS, and IAM; log ingestion and auditing, S3 object monitoring, and a Kubernetes agent that certifies each deployment runs the expected image.
- Built an event-driven Java/Quarkus pipeline for binary metering files (FTPS/S3, NDJSON decoding, idempotent persistence, lease-based retries) and rewrote it in C++20 (~30k lines) keeping every contract intact.
- Migrated about eight WPF and WinForms desktop apps to Avalonia on .NET 8/10: one codebase for Windows, Linux, macOS, Android, and iOS, with Clean Architecture, dependency injection, and mock infrastructure for hardware-free testing. Each migration took about 2–3 weeks, and I left the team a dotnet new template.
- Built a cross-platform app to manage BLE devices: mesh provisioning, a CBOR command protocol, firmware-over-the-air (FOTA) updates, metering-file decoding, and dashboards with charts and maps, backed by about 400 automated tests in GitLab CI.
- Shipped a Flutter field-operations app to Android and iOS: offline login with an encrypted cache (ChaCha20), a sync queue, GPS/GNSS capture, barcode scanning, and gRPC over TLS.
- Created a Go service that estimates the 6- and 24-hour probability of long outages per grid device from open weather and seismic data, with its own web dashboard.

Technologies: AWS, SQS, S3, Go, NATS, Kubernetes, Azure, Docker, C#, F#, Avalonia, C++20, Java/Quarkus, Flutter, TypeScript, React, PostgreSQL, SQL Server

Sr .NET Developer

IMPI (Mexican Institute of Industrial Property) Apr 2025 – Jun 2025

- Built .NET C# WinForms applications and REST APIs, including multithreaded processing.
- Administered Linux servers, ran backups and technology benchmarks, and managed repositories in Azure DevOps.

Technologies: C#, .NET, WinForms, Azure DevOps, Linux

Robotics and AI Researcher

ALIROB, Instituto Politécnico Nacional Jan 2023 – Jan 2025

- Built cellular automata simulations in C# (.NET 6 and 8) applied to artificial life and natural computing.
- Designed and validated bio-inspired models based on Physarum polycephalum for autonomous navigation, with a C++ simulator and a Raspberry Pi robot.

Technologies: C#, C++, SFML, Raspberry Pi, Java, JavaScript

PUBLICATIONS AND AWARDS

- **International publication — CANDAR 2024** (November 2024). ASON workshop, International Symposium on Computing and Networking, Okinawa, Japan. http://bncss.org/index.php/bncss/article/view/183;
- **Graduated with honors** (December 2024). Escuela Superior de Cómputo, IPN.
- **Finalist — Talent Land Hackathon 2022** (2022). Talent Land México.

EDUCATION

B.Sc. in Computer Systems Engineering

Escuela Superior de Cómputo (ESCOM), Instituto Politécnico Nacional 2020 – 2024

Graduated with honors by thesis (December 2024). Thesis: "Design of an autonomous monitoring system based on cellular automata and Physarum polycephalum."

Programming Technician (high school)

CECyT 9 «Juan de Dios Bátiz», Instituto Politécnico Nacional 2017 – 2020

TECHNICAL SKILLS

Programming languages: Go, C#, F#, C/C++, Java, TypeScript, JavaScript, Dart.

Backend: NATS, REST / OpenAPI, gRPC, Quarkus, ASP.NET Core, Node.js, JWT.

Cloud (AWS): SQS, S3, EC2, EKS, CloudTrail, EventBridge, DynamoDB, IAM / STS.

Infrastructure: Kubernetes, Kustomize, Docker, Azure, GitLab CI, GitHub Actions, Linux, Git.

Databases: PostgreSQL, SQL Server, DynamoDB, SQLite.

Desktop & mobile: Avalonia, WPF, WinForms, Flutter, Android, iOS.

Frontend: React, Vite, Module Federation, Blazor WebAssembly, Bolero.

Protocols & IoT: BLE / mesh, CBOR, FOTA, DNP3, USB-serial, Web Serial, TCP/UDP, FTPS.

Architecture & quality: Clean Architecture, MVU / Elmish, MVVM, Microservices, xUnit, Contract tests, ADRs.

Systems & networking: Windows Server, DNS, Firewalls, Grafana, PowerShell, Bash.

Also used: Python, Lua, VHDL, Assembly, GraphQL, Express, MySQL, Oracle DB, JavaFX, SFML, PHP, JSP, Azure DevOps.

LANGUAGES

Spanish: Native · English: Upper intermediate (B2) · Japanese: Basic (A2) · Chinese: Basic (A1)

Updated: November 2026. Full version with case studies and screenshots: https://isgarcia-zero.github.io/